EX-11 Write the Python program for Map Coloring using Constraint Satisfaction Problem (CSP)

AIM

To develop a Python program to solve the **Map Coloring Problem** using the **Constraint Satisfaction Problem (CSP)** approach, ensuring that no two adjacent regions have the same color.

ALGORITHM

Step 1: Start the program.

Step 2: Define the map as a graph where each region is represented as a node and neighboring regions are connected by edges.

Step 3: Define the available colors for coloring the regions.

Step 4: Assign a color to the first region.

Step 5: Check whether the assigned color satisfies the constraint that no adjacent regions have the same color.

Step 6: If the color is valid, assign colors to the remaining regions recursively.

Step 7: If no valid color is available for a region, backtrack and assign a different color to the previous region.

Step 8: Repeat the process until all regions are colored successfully.

Step 9: Display the color assigned to each region.

Step 10: Stop the program.

PROGRAM CODE

```
graph = {
    'A': ['B', 'C'],
    'B': ['A', 'C', 'D'],
    'C': ['A', 'B', 'D'],
    'D': ['B', 'C']
```

```
}

colors = ['Red', 'Green', 'Blue']

result = {}

def is_safe(node, color):
    for neighbour in graph[node]:
        if neighbour in result and result[neighbour] == color:
            return False
    return True

def solve(nodes, index):
    if index == len(nodes):
        return True

    node = nodes[index]

    for color in colors:
        if is_safe(node, color):
            result[node] = color

            if solve(nodes, index + 1):
                return True
```

```
            del result[node]

    return False

nodes = list(graph.keys())

if solve(nodes, 0):
    print("\nMap Coloring Solution:")
    for node in nodes:
        print(node, "=", result[node])
else:
    print("No solution exists")
```

OUTPUT

```
Map Coloring Solution:
A = Red
B = Green
C = Blue
D = Red
```

RESULT

The Python program successfully implements the **Map Coloring Problem** using the **Constraint Satisfaction Problem (CSP)** approach and assigns colors to all regions such that no two adjacent regions have the same color.